367352(67)

B. E. (Third Semester) Examination, Nov.-Dec. 2015

CSVTUonline.com

(New Scheme)

(Mechatronics Branch)

METROLOGY and INSTRUMENTATION

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

CSVTUonline.com

Note : *Attempt all questions. Part (a) from each question is compulsory. Attempt any two parts from (b), (c) and (d) of each question. Draw neat diagrams wherever necessary.*

1. (a) What is meant by a Transducer? What are the different types of transducer? Gives names only. 2

(ii) Roundness

(iii) Flatness

(iv) Straightness

(c) Explain the use of gauge blocks. What is a comparator? Explain different types of comparators. 7

(d) Explain the working of autocollimator? What are the applications of autocollimator? 7

CSVTUonline.com

5. (a) Write uses of the following : 2

(i) Interferometer

(ii) Comparator

(b) Explain principle and use of interferometer with suitable figure. 7

(c) What is Coordinate Measuring Machine? Explain the constructional features and its applications. 7

(d) Explain the two wire method for screw threads effective diameter measurement. 7

CSVTUonline.com

(b) Explain the generalised measurement system with the help of diagram. Discuss its functional elements. 7

(c) The following readings are taken of a certain physical length with the help of a micrometer screw :

1·41, 1·45, 1·63, 1·54, 1·49, 1·51, 1·60, 1·55, 1·47, 1·65 mm

Assuming that only random errors are present, calculate the arithmetic mean, the average deviation, standard deviation, variance and the probable error of the reading. 7

CSVTUonline.com

(d) Explain clearly the factors to be taken into considerations while selecting an instrument for particular measurement application. 7

2. (a) Define Strain Gauge. List points needed for consideration while designing strain gauge. 2

(b) Describe the factors which influence response of a temperature sensing device. What are the errors that may rise in the measurement of temperature? Discuss briefly. 7

CSVTUonline.com

(c) Discuss the classification of low pressure gauges in brief. Explain the Mcleod Guage with the help of neat diagram. 7

(d) What do you mean by temperature compensation in strain gauges? How temperature compensation is achieved with the help of Poisson's method. 7

3. (a) What is meant by Laminar Flow and Turbuleent Flow? Give equation of Reynold's Number. 2

(b) Explain the principle of operation of : 7

(i) Hot wire anemometer

(ii) Thermal flow meter

CSVTUonline.com

(c) What do you understand by Data Transmission? Name the methods adopted for data transmission and describe operating principle of hydraulic and pneumatic transmission systems. 7

(d) Give the brief account of Venturi flow meter with the help of diagram. Explain its advantages and limitations. 7

4. (a) Define measurements. What is primary, secondary and tertiary measurement? 2

(b) Explain the following terms with suitable engineering examples : 7

(i) Circularity